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#. Solve Q3 of PS1 when both θ, σ^2 are unknown.

* Method of Scoring for computing MLE:

$X_1, \dots, X_n \overset{iid}{\sim} f(x|\theta)$

~~log~~ Log-likelihood $= \sum_{i=1}^{n} \log\{f(X_i|\theta)\}$

$\Rightarrow$ Maximum-Likelihood equation: $\sum_{i=1}^{n} \frac{f'(X_i|\theta)}{f(X_i|\theta)} = 0$
(ML)

For Newton-Raphson (NR) iteration we use 2nd derivative (wrt. θ): $\sum_{i=1}^{n} \left[\frac{f''(X_i|\theta)}{f(X_i|\theta)} - \left\{ \frac{f'(X_i|\theta)}{f(X_i|\theta)} \right\}^2 \right]$

Replace $\longrightarrow$ By its expectation = Fisher Information
(Indp. of θ when θ is location paramter)

$\rightarrow$ NF factoriz^n:

Likelihood $= h(X) \cdot g(\theta, T(X))$

Hence MLE is a func^n of suff. stat. $T(X)$.

~~If $\hat{\theta}$ is Let $\hat{\theta}$ be~~

- Let ψ be a func^n of θ:
- If $\hat{\theta}$ is unb. est. of θ, then $\psi(\hat{\theta})$ is unb. est. of $\psi(\theta)$ surely when ψ is linear. Same for Baye's estimator.
- For MLE, ψ can be any one-one function for us to be sure, about that.

$\rightarrow$ $X \sim P_\theta$ $\quad \{P_\theta\}_{\theta \in \Theta}$

$T(X) = T$ is an est. of θ.

$R(\theta, T) = E_\theta[(T-\theta)^2]$

$\underset{\theta \in \Theta}{(\sup)\max} R(\theta, T) =$ Maximum risk of T

Def^n: An estimator T^* is called a Minimax estimator if

$$\sup_{\theta \in \Theta} R(\theta, T^*) = \inf_T \sup_{\theta \in \Theta} R(\theta, T)$$

Ex1: $X \sim Bin(n, \theta)$, $0 < \theta < 1$

$$\pi(\theta) = \frac{1}{\beta(p,q)} \theta^{p-1} (1-\theta)^{q-1} \qquad p, q > 0$$

Baye's Estimate $= \dfrac{p+X}{p+q+n}$ (# How)

$$UMVUE = \frac{X}{n} = T \text{ (say)}$$

We know $\max_{\theta \in \Theta} R(\theta, T) = \dfrac{1}{4n}$ (# Easy to see)

- Result: $X \sim P_\theta$, $\theta \in \Theta$, $\pi(\theta)$.
  T_π is the Baye's estimate
  $R(\theta, T_\pi) =$ Constant free of θ
  Then, T_π is a minimax est. of θ.

Proof: $\sup_{\theta \in \Theta} R(\theta, T_\pi) = c = r(\pi, T_\pi) \le r(\pi, T)$

$\le \sup_{\theta \in \Theta} R(\theta, T)$ //

(smth. smth. baye's risk) (# What?)

• In Ex 1, the Baye's estimate w/ beta prior s.t. $p = q = \frac{\sqrt{n}}{2}$ is a minimax estimator.

A Baye's estimate (generalized) under improper prior and they are equalizer estimates, then, can they be minimax ?